

Better Flashcards

Online learning platform based on flashcards method and spiced with spaced repetition

HEIG-VD WEB 2026

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10.06.2026

Inspiration source (for features)

Anki

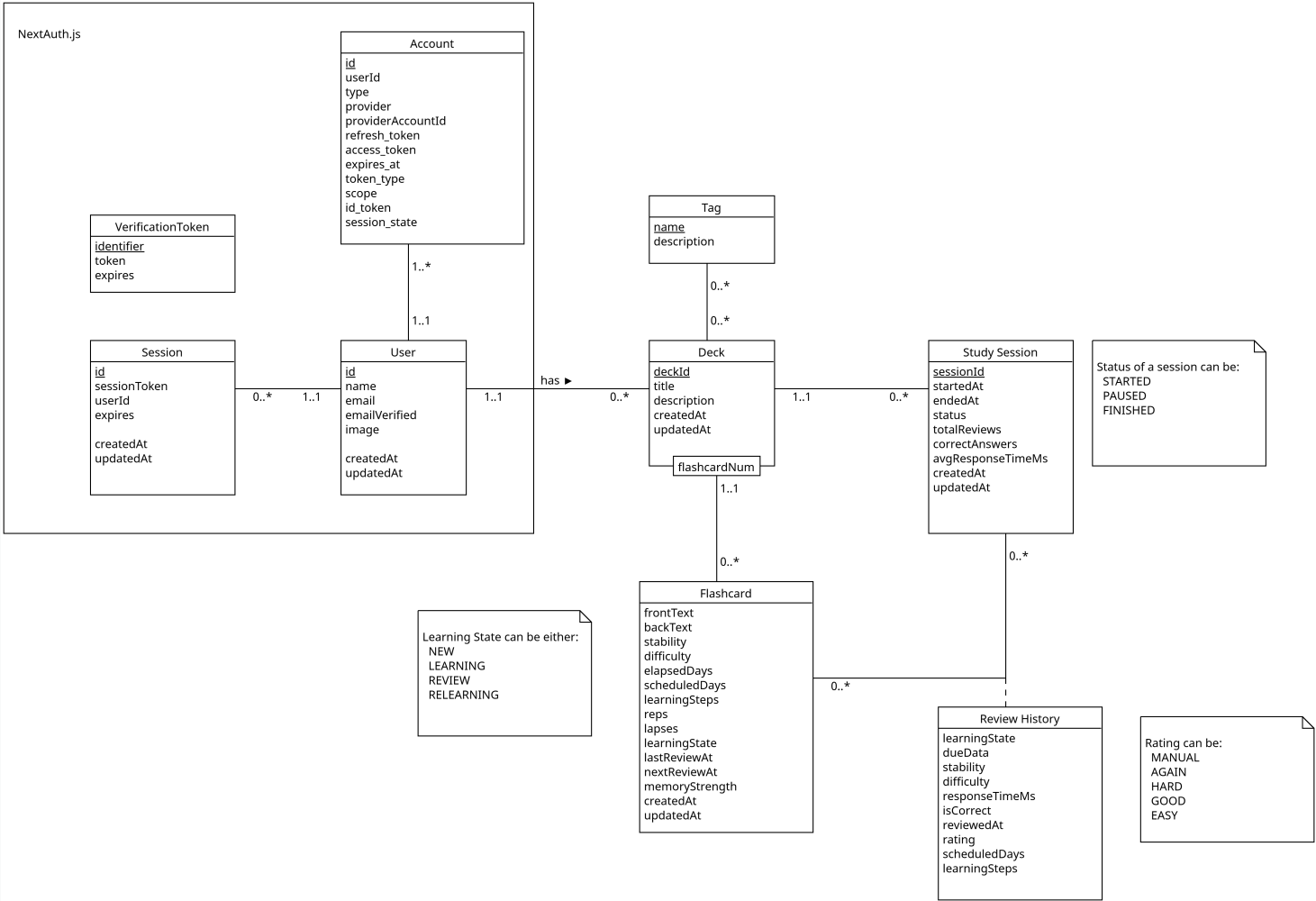
Objectives to Features

Objectives

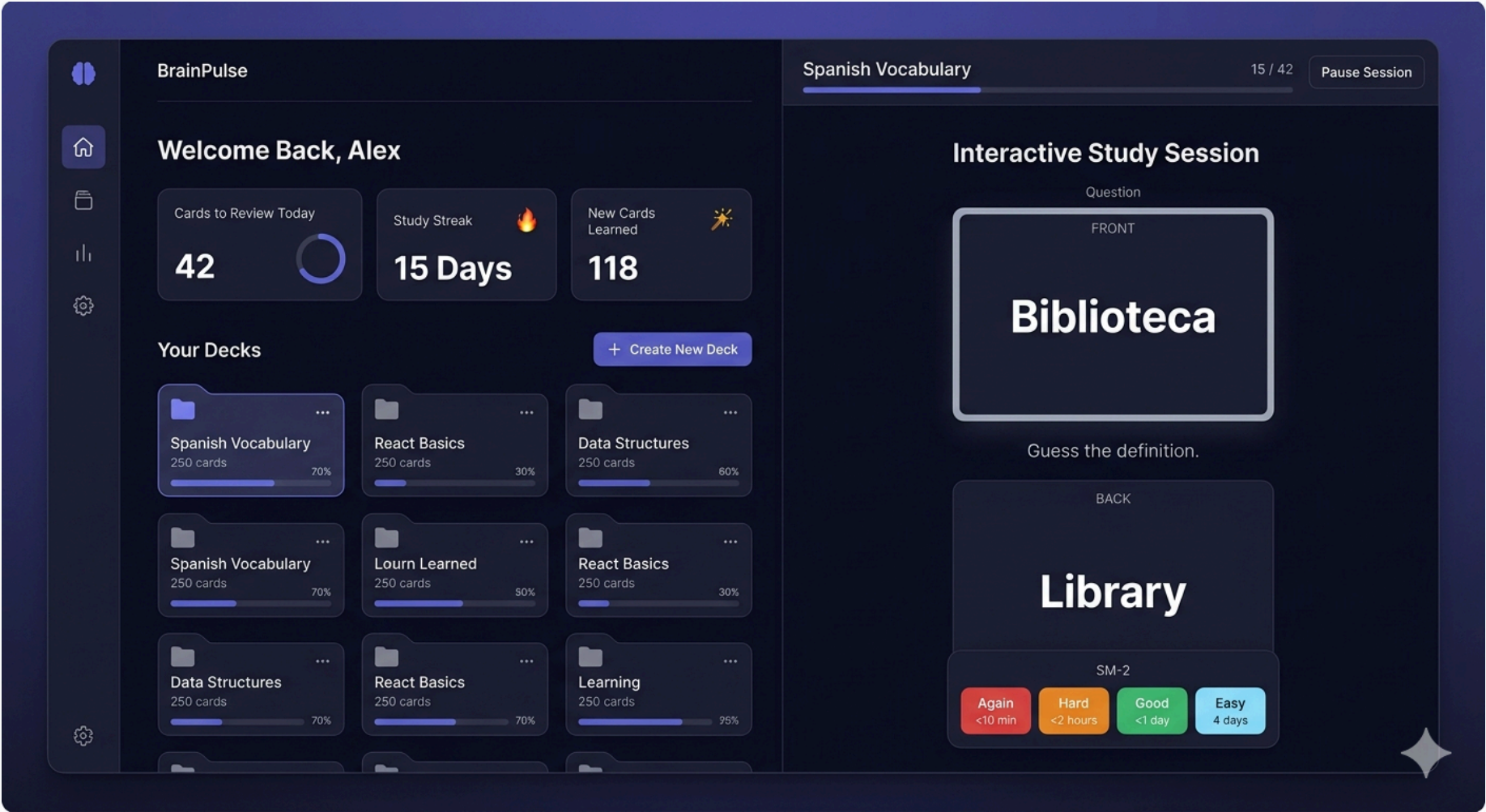
- Create and organize personalized learning materials into structured study decks
- Improve long-term knowledge retention through adaptive repetition and review scheduling
- Track learning progress and identify weak areas over time

Core Features

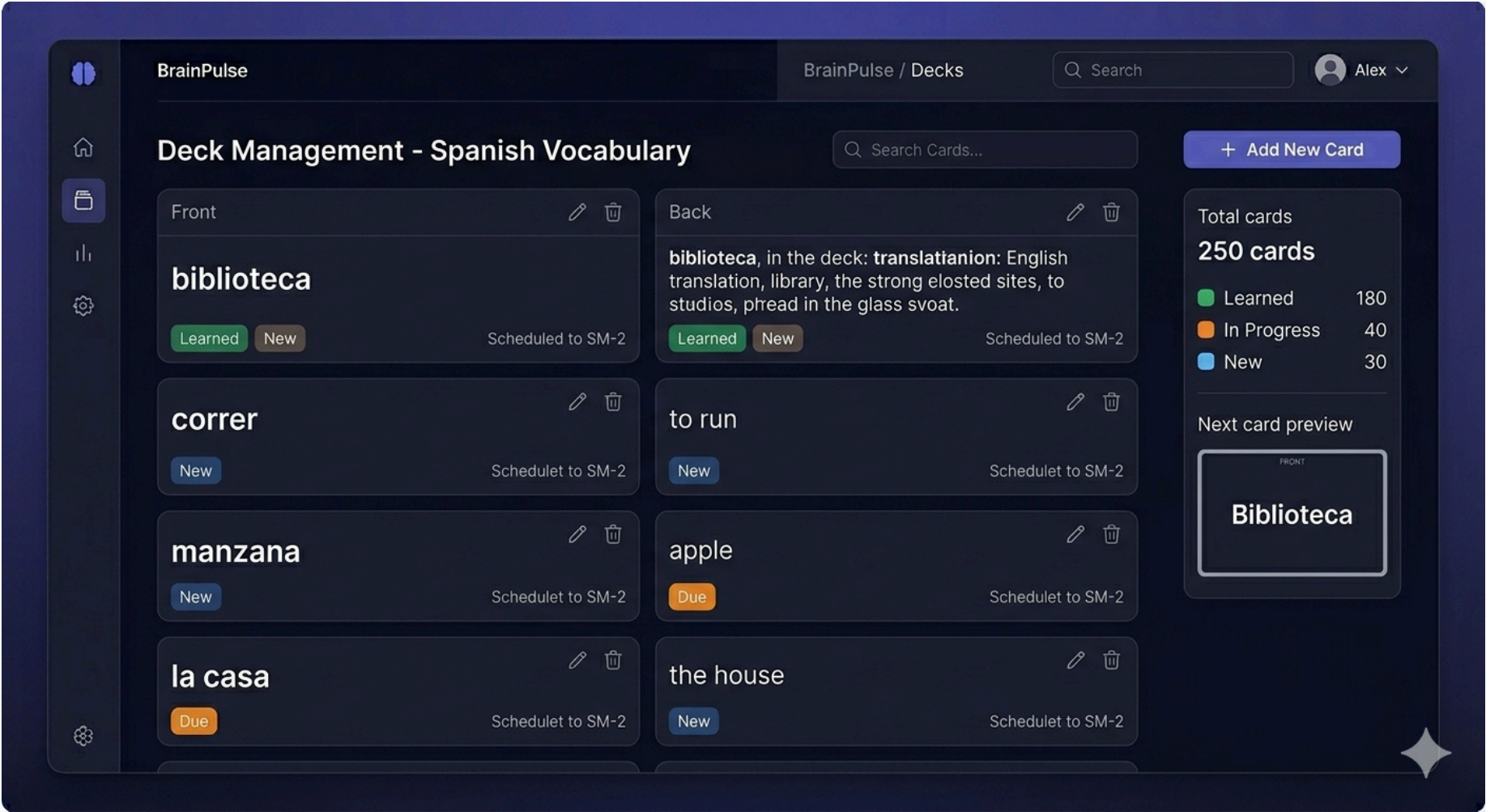
Data Model



Generated mockups (Gemini - 1)



Generated mockups (Gemini - 2)



Generated mockups (ChatGPT - 1)

BrainPulse

Dashboard

Decks

Study

Analytics

Settings

Decks > Spanish Vocabulary

Search cards...

Spanish Vocabulary

Essential Spanish words and phrases for everyday conversations.

150

Total Cards

72

Learned

42

In Progress

36

New

Created
May 10, 2024

All Cards

Learned

In Progress

New

Search in deck...

All Status

All Tags

Sort by: Newest

#	Front (Spanish)	Back (English)	Status	Tags	Last Reviewed	Actions
1	hola	hello	Learned	Greetings	2 days ago	
2	gracias	thank you	Learned	Expressions	1 day ago	
3	por favor	please	In Progress	Expressions	3 days ago	
4	adiós	goodbye	In Progress	Greetings	1 week ago	
5	casa	house	New	Nouns	—	
6	comer	to eat	New	Verbs	—	
7	agua	water	New	Nouns	—	
8	amigo	friend	New	Nouns	—	

Showing 1 to 8 of 150 cards

<

1

2

3

...

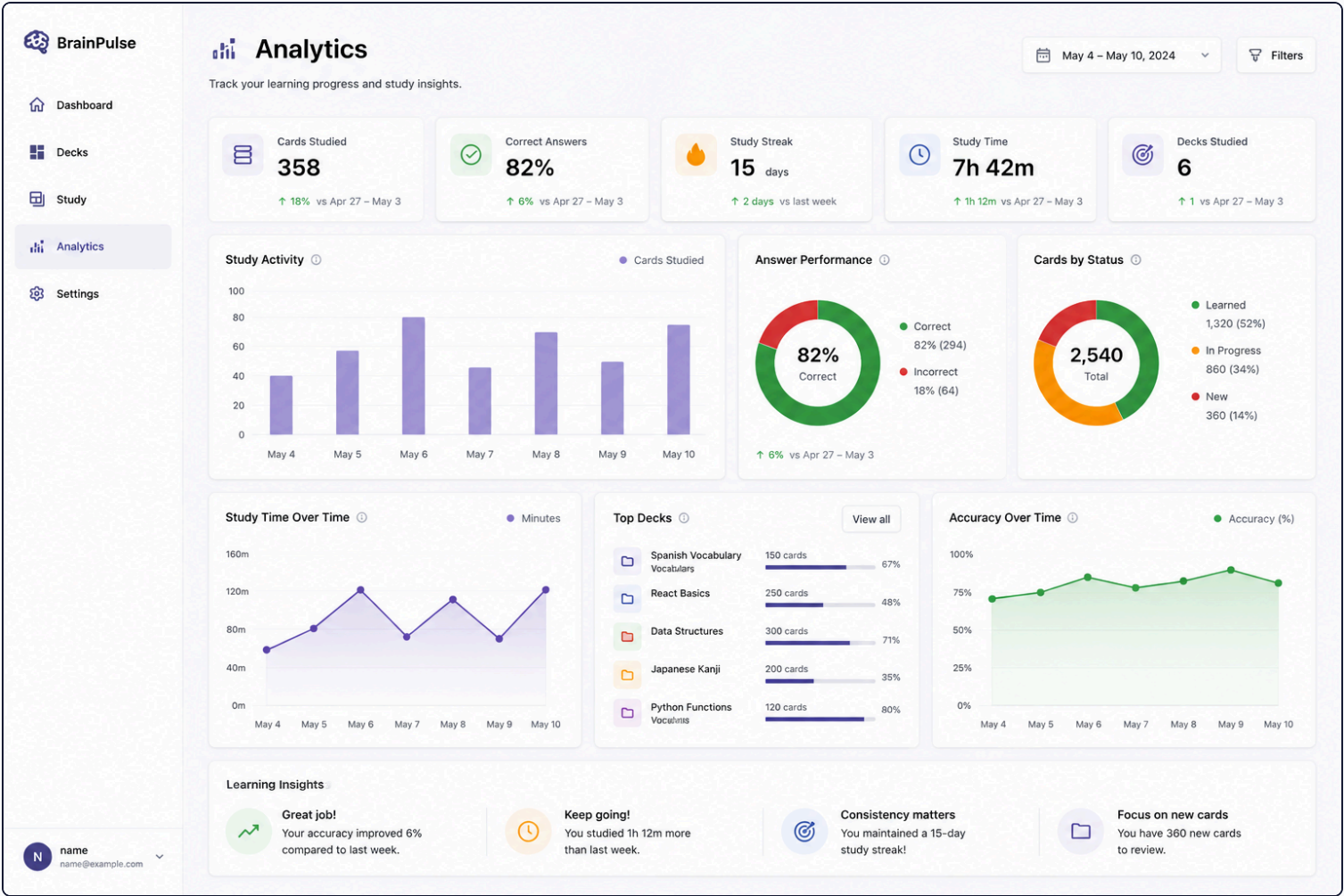
19

>

N name

name@example.com

Generated mockups (ChatGPT - 2)



Technologies selection

Initially

- Next.js
- MUI
- Zustand
- Motion React
- Prisma
- Sqlite
- NextAuth.js
- Typescript
- SuperMemo 2 (SM-2)

Currently

- Next.js
- MUI
- ~~Motion React~~
- Zustand
- Prisma
- Sqlite
- NextAuth.js
- Typescript
- ~~SM-2~~
- SWR (*new*)
- Zod (*new*)
- Free Spaced Repetition Scheduler (*new*)
- Date-fns (*new*)

Challenges (1)

How to type this whole thing ?

- Use Prisma types generated from models ? Is it enough ?

```
model StudySession {  
  sessionId String @id @default(cuid())  
  startedAt DateTime @default(now())  
  endedAt DateTime?  
  status StudySessionStatusEnum  
  totalReviews Int @default(0)  
  correctAnswers Int @default(0)  
  avgResponseTimeMs Int?  
  deckId String  
  deck Deck @relation(fields: [deckId], references: [deckId])  
  createdAt DateTime @default(now())  
  updatedAt DateTime @updatedAt  
  reviewedCards ReviewHistory[]  
}
```

- Not really

```
const session: StudySessionModel = await fetch(  
  '/api/session/start',  
  {  
    method: 'POST',  
    body: JSON.stringify({ deckId: selectedDeck?.deckId }),  
  }).then(res => {  
    if (!res.ok) {  
      throw new Error(`Failed to create new session for deck with id=${selectedDeck?.deckId}`)  
    }  
    return res.json()  
  })  
// session.startedAt is string and not a Date  
// session.status is string and not enum
```

Demo

Respect of defined specification

Further development

Thank you for attention !
